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Transpose

Transpose in Excel is used to change to orientation of data from vertical to horizontal and Horizontal to vertical

Ex

A

1. FRUITS

2. Apple

3. Mango

4. Banana

After transpose

A

B

C

D

FRUITS

Apple

Mango

Banana

Transpose using past special

- Select the data in A column till A4
- Press ctrl + c
- Select the destination cell
- click on paste special
- Select Transpose
- click OK

Again if we want to convert horizontal to vertical select the cells and copy select

destination cell then paste special, select transpose, click ok

Paste Link

Paste link creates a connection b/w the copied data and the original data. If any changes are made in original data will reflect in the copied data

How to paste link

Consider that I have data from A1 to A10

- Select this column till A10
- then ctrl + c
- Select the destination cell
- click on paste special
- Select paste link
- click OK

Paste special Operations

Paste special is not only for transpose. It can also perform mathematical operations while pasting.

~~The main o~~

The main operations are

- Add
- Subtract
- Multiply
- Divide

Ex: Suppose my cells contains:

	A	B	C	D	E
	Data	ADD	Sub	Multiply	Div
1	16	34	5	62	7
2	54	7	43	23	8
3	28	89	21	14	34
4	34	32	67	27	52

These are the data and I should add the data with B column and get the answer in same column and also same with the other three operations also

How to do these operations

- Select data from A2 to A5
- click ctrl + c
- then select B2 cell and then click on paste special.

References

A reference means the address

- Select add option and click OK
- drag it till B5 cell

The same procedure to all those 3 operations also

The final results will appear like this

	A	B	C	D	E
1	Data	Add	Sub	Multiply	Div
2	16	50	-11	992	0.4375
3	54	61	-11	1242	0.148148
4	28	117	-7	392	1.214286
5	34	66	33	918	1.529412

- Here the B column is added with A
- C column is subtracted with A column
So it shows as a negative value
- E column is divided with A column

$$A2 = 160$$

If I want to convert my values into negative $(= A2 * -1)$
this becomes -160 in B2

The convert this to positive

$$(= B2 * -1)$$

this becomes 160

references

A reference means the address of a cell that we use in a formula

Ex: $=A1 + B1$

There are 2 types of cell references

i Relative reference

A relative reference changes automatically when we copy the formula to another cell.

Syntax: Column Row

Ex: A1

Suppose:	A	B	C
	10	2	$=A1 * B1$ (20)
	20	3	60

In C1 result = 20, Now copy C1 to C2

Excel automatically changes ($=A2 * B2$)
The results become 60

ii Absolute reference

This does not change when we copy the formula we use the (\$) symbol

Syntax: \$ column \$ row

Ex: $=\$A\1

Ex:	A	B	C	D
	Sana	15,000	$= B1 * \$D\1	8%
	Maan	22,000		

The absolute ref is a cell reference that remains fixed when a formula is copied or moved to another cell

Here column D is fixed (\$D)

Row 1 is fixed (\$1)

So \$D\$1 will always refer to cell D1

So if we drag down the formula will remain fixed to D1 only the B column cells changes

Mixed Reference

This is a combination of relative reference and absolute reference

\$A1 → A is fixed column, 1 changes row wise

A\$1 → A is changing column, 1 is fixed row

\$A\$1 → A is fixed column, 1 is fixed row

A1 → A changes, 1 row also changes

Ex:	A	B	C	D
1	1	2	3	4
2	2	= \$A2 * B\$1	6	8
3	3	6	9	12
4	4	8	12	16

Here we want to multiply as a matrix
 So here ($= \$A2 * B\1) A column is fixed and
 2 will change as 3, 4, 5
 B will change as C, D, E, F... and the cell
 2 will be fixed

Round, Round up, Round down functions

i Round 2 decimal point

If we want 2 decimal place, Excel looks
 at the 3rd decimal digit

Syntax: = ROUND (number, num-digit)

Ex: 4.8319 is in (A2) = ROUND (A2, 2)

o/p : 4.83

Ex: 6.12 is in (A7) = ROUND (A7, 2)

o/p : 6.12

ii) **POWER()**: is used to raise a no. to a specific power

Syntax: $=POWER(\text{number}, \text{power})$

Ex: $=POWER(G2, H2)$ ($G2=4$, $H2=2$)

O/P: 16

or else ($=G2^H2$) we can use power (^) operator

iii) **ROMAN()**: converts the normal number into a roman numeral

Syntax: $=ROMAN(\text{number})$

Ex: $=ROMAN(L2)$ ($L2=4$)

O/P: IV

iv) **ARABIC()**: converts roman numeral into a arabic number that is normal no.

Syntax: $=ARABIC(\text{text})$

Ex: $=ARABIC(M2)$ ($M2=IV$)

O/P: 4

v) **REPEAT()**: REPT() function repeats a given text or characters a specified no. of times

Syntax: = REPT (text, number - times)

Ex: = REPT (P2, 5) (P2 = 4)

O/p: 44444

I want to repeat the number according to its value

Syntax: = REPT (text, num - times)

Ex: = REPT (P2, P2) (P2 = 4)

O/p: 4444

= REPT (P3, P3) (P3 = 6)

O/p: 666666

other method = REPT (P2, &P2)

Here, P column is fixed, 2 will change to 3, 4, 5

vi SQRT(): used to find the square root of a no. A square root is a no. that, when multiplied by itself, gives the original no.

Ex: $5 \times 5 = 25$, so the SQRT of 25 is 5

Syntax : $= \text{SQRT} (\text{number})$

Ex : $= \text{SQRT} (V2)$ ($V2 = 16$)

O/p : 4

How to convert positive values to ~~neg~~ negative

$V2 = 16$

Ex : $= V2^* - 1$

O/p : -16

To convert to positive

Ex : $= \text{ABS} (X2)$ ($X2 = -16$)

O/p : 16

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Employee Data

A = Emp code

B = Emp name

C = Address

D = city

E = Region

F = Department

G = Basic salary

- 1 Find the total basic salary

Sum () : used to calculate the total of numerical values

Syntax : = Sum (number 1, ~~Ex~~ [number 2])

Ex : = Sum (G2 : G21)

O/p : 1,84,600.00

- 2 Find Average salary

Average () : used to calculate the average value of numbers

Syntax : = AVERAGE (range), = AVERAGE (num1, [num2])

Ex : = AVERAGE (G2 : G12)

O/p : 9,230.00

- 3 Maximum salary

Max () function

Syntax : = Max (num 1, (num 2))

Ex : = MAX (G2 : G21)

O/p : 16,250

- 4 Minimum Salary

Syntax : = Min (num 1, (num 2))

Ex : = Min (G2 : G21)

O/p : 4,500

5 Count no. of salary
count(): counts no. of cells containing no's
Syntax := count (value1 [value2])
Ex := count (G2:G21)
o/p: 20

6 Count num of employees by name
counta(): counts the no. of non empty cells
Syntax := counta (value1 [value2])
Ex := counta (B2:B12)
o/p: 20

7 Find second largest salary
large(): returns the n^{th} largest value
from a range
Syntax := LARGE (array, k)
Ex := LARGE (G2:G21, 2)
o/p: 16,250

also these function accepts duplicate value
if 16,250 is repeated 2 times the 3rd
largest salary will be shown as 16,250
so other method for this is

Ex: =LARGE (unique (G2:G21), 2)

o/p: 16,250

8 Find 3rd smallest salary

Small () = finds and returns the Nth smallest value from a range

Syntax = SMALL (range, k)

Ex: =SMALL (G2:G21, 3)

o/p: 6275

other method to avoid duplicate

Ex: =SMALL (unique (G2:G21), 3)

o/p: 6275

unique (): returns only unique values by removing duplicates

Syntax: =unique (array, [by-col], [exactly-once])

Ex: =unique (G2:G21)

returns only unique values

9 Find total salary of marketing team

Sumif () : adds value based on one condition

Syntax: =SUMIF (range, criteria, [sum-range])

Ex: =SUMIF (F2:F21, F12, G2:G21), =SUMIF (F2:F21, "marketing", G2:G21)

o/p: 31,500

- 10 Find total salary of employee who are earning 10,000

Ex: =SUMIF(G2:G21, G2), =SUMIF(G2:G21, 10,000)

o/p: 30,000

- 11 Find total salary of employee who are earning above 10,000

Ex: =SUMIF(G2:G21, ">10,000", G2:G21)

o/p: 90250

- 12 Find average salary of west region employees

Ex: =AVERAGEIF(E2:E21, E10, G2:G21)

Syntax: AVERAGEIF (range, criteria, [average-range])

o/p: 8500

Ex: =AVERAGEIF(E2:E21, "w", G2:G21)

- 13 Find maximum salary in R&D department
=maxifs [max-range, criteria-range1, criteria1, [criteria-range2, criteria2]

Ex: =maxifs (G2:G21, F2:F21, "R&D")

=maxifs[G2:G21,P2:P21,FS)

o/p :12,000